

Data Analysis with R and Python

Semestral Examination

23 MAY 2025, 10:30AM – 1:30PM

Bachelor of Statistical Data Science

TOTAL POINTS: 50

INSTRUCTIONS

1. Write your answers on the question paper. You may use separate sheets for calculations, but submit only the filled in question paper.
2. Keep your answers brief and to the point. Write only the answers to the questions asked.
3. Do not write any R code or any other explanation unless instructed.
4. This is a practical examination you may use laptops or desktops with R and necessary packages pre-installed, as per instructions given earlier.
5. However, you are not allowed to access the internet during the examination.

NAME (IN CAPITAL LETTERS): _____

ROLL NUMBER: _____

Question 1

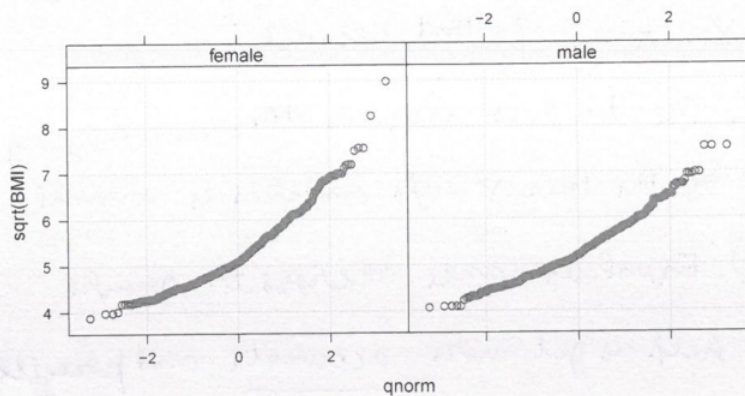
Consider the NHANES dataset contained in the NHANES R package, and in particular the variables `Age`, `Gender`, `Race1`, and `BMI`. We are interested in studying whether the population average of BMI for White males and White females are significantly different among people between the ages of 20 and 40 years, inclusive.

Obtain the appropriate subset of the NHANES data, and use the `t.test()` function to perform a suitable two-sample t-test assuming equal variance.

(a) What is the value of the test statistic? Is the test for equality of means rejected at level 0.01? Briefly describe how you arrived at your conclusion.

(b) What is the 80% confidence interval for the difference between the two means, as computed by the `t.test()` call above? Give your answer rounded to three places after the decimal point. [6]

2. Another statistician suggests that instead of the raw BMI values, it would have been more appropriate to use the transformed variable $\sqrt{\text{BMI}}$ to test for equality of population means in question 1. As justification, he provides the following Q-Q plot of the **transformed** data.



Using R or Python, obtain the corresponding Q-Q plot of the untransformed BMI (you are *not* required to show your code). Compare the two Q-Q plots, and based on what you observe, answer the following questions.

- (a) For which case (original vs transformed) is the assumption of Normality more reasonable? [3]
Briefly justify your answer.

Question 2

Another statistician suggests that instead of the raw BMI values, it would have been more appropriate to use the transformed variable $\sqrt{\text{BMI}}$ to test for equality of population means in question 1. As justification, he provides the following Q-Q plot of the transformed data.

Using R or Python, obtain the corresponding Q-Q plot of the untransformed BMI (you are not required to show your code). Compare the two Q-Q plots, and based on what you observe, answer the following questions.

(a) For which case (original vs transformed) is the assumption of Normality more reasonable? [3] Briefly justify your answer.

(b) For which case is the assumption of equal variance more reasonable? Briefly justify your answer. [3]

(c) Is the assumption of Normality important for the two-sample t-test used in question 1? Briefly justify your answer. [4]

Question 3

A multiple choice test has thirty questions. Each question has four options, of which exactly one is correct. A candidate passes the test if they answer at least 12 questions correctly. There is no penalty for incorrect answers.

(a) A candidate chooses one of the four options randomly as their answer for all thirty questions. [4] Let p_1 be the probability that the candidate will pass. Calculate and enter the exact value of p_1 rounded to five places after the decimal point.

(b) Another candidate answers the test as follows.

For the first ten questions, the candidate is able to eliminate exactly two options as incorrect, and randomly chooses one of the two remaining options as their answer.

For the next ten questions, the candidate is able to eliminate exactly one incorrect answer, and randomly chooses one of the three remaining options as their answer.

For the last ten questions, the candidate is unable to eliminate any answer, and randomly chooses one of the four options as their answer.

Let p_2 be the probability that the candidate will pass. Estimate p_2 by simulating the procedure described above 10000 times, and enter the number of successes (i.e., the number of cases in which the candidate passes) you observe. [6]

Question 4

Suppose the n -tuple of observations $(X_1, X_2, \dots, X_n)$ arises from the following random process, for some unknown $\rho \in \mathbb{R}$:

$$\begin{aligned} X_1 &\sim N(0, 1) \\ \text{Given } X_1 = x_1, \quad X_2 &\sim N(\rho x_1, 1) \\ \text{Given } X_2 = x_2, \quad X_3 &\sim N(\rho x_2, 1) \\ &\vdots \\ \text{Given } X_{n-1} = x_{n-1}, \quad X_n &\sim N(\rho x_{n-1}, 1) \end{aligned}$$

The joint density of $(X_1, X_2, \dots, X_n)$ is then given by

$$\begin{aligned} f_\rho(x_1, x_2, \dots, x_n) &= f(x_1)f(x_2|x_1)f(x_3|x_2)\dots f(x_n|x_{n-1}) \\ &= \phi_{0,1}(x_1)\phi_{\rho x_1,1}(x_2)\phi_{\rho x_2,1}(x_3)\dots \phi_{\rho x_{n-1},1}(x_n) \end{aligned}$$

Here, $\phi_{\mu,\sigma^2}(x)$ denotes the density of the $N(\mu, \sigma^2)$ distribution evaluated at x .

Using the above result, which you may assume to be true, find the maximum likelihood estimator of ρ for this model, based on the following simulated data.

```
set.seed(20250523)
n <- 100; rho <- 0
x <- numeric(n)
x[1] <- rnorm(1, mean=0)
for (i in 2:n) x[i] <- rnorm(1, mean = rho * x[i-1])
str(x)
## num [1:100] 1.945 -0.543 1.525 -1.19 1.352 ...
```

(a) Give your answer rounded to three places after the decimal point. [15]

(b) Briefly describe the approach you used to find your answer (no code or calculations required). [5]

Good luck!